Basic Principles of Medicine 1

Module: Foundation to Medicine (FTM) | DLA 21 before FTM LEC 38

Anatomy of the Autonomic Nervous System

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Objectives

SOM.MKII.BPM1.1.FTM.1.ANAT.1009	Describe the components of the autonomic nervous system.
SOM.MKII.BPM1.1.FTM.1.ANAT.1013	List the three target tissues of visceromotor (ANS) fibers.
SOM.MKII.BPM1.1.FTM.1.ANAT.1014	Define the terms preganglionic and postganglionic.
SOM.MKII.BPM1.1.FTM.1.ANAT.1015	List the four cranial nerves that contain preganglionic parasympathetic fibers.
SOM.MKII.BPM1.1.FTM.1.ANAT.1016	List the segments of the spinal cord that contain the neural cell bodies of preganglionic parasympathetic fibers.
SOM.MKII.BPM1.1.FTM.1.ANAT.1017	List the locations of the neural cell bodies of the postganglionic parasympathetic fibers
SOM.MKII.BPM1.1.FTM.1.ANAT.1018	Identify the segments of the spinal cord that contain the neural cell bodies of preganglionic sympathetic fibers and use this information to interpret clinical signs and symptoms
SOM.MKII.BPM1.1.FTM.1.ANAT.1019	List the locations of the neural cell bodies of the postganglionic sympathetic fibers
SOM.MKII.BPM1.1.FTM.1.ANAT.1020	Describe the location and extent of the sympathetic trunk.

Terminology

- Ganglion/ganglia
- Nucleus/nuclei
- Somatic
- Visceral
- Autonomic
- Sensory
- Motor
- Afferent
- Efferent
- Peripheral nervous system (PNS)
- Central nervous system (CNS)

Review: Nervous System Integration

Somatic

Receptors in **Body Wall** for conscious sensation:
pain, pressure, touch, temperature, propiocception

somatic sensory/afferent

somatic motor/efferent

Somatic

Voluntary control of **Skeletal Muscle**

PNS

CNS

PNS

Visceral

Receptors in **Organs** and **Tissues**
Distension,
Inflammation,
Ischemia,
Physiological changes,
Abnormal muscle spasm

visceral sensory/afferent

visceral motor/efferent (AUTONOMIC)

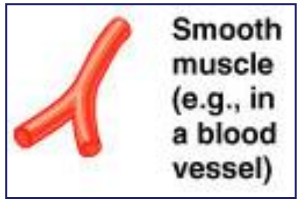
Visceral

- Involuntary control of
- **Smooth Muscle**
- **Cardiac Muscle**
- **Glands**

General Characteristics of ANS

Visceral Motor or **Efferent** system - innervates **3** target tissues:

•Smooth muscle (eg. GI tract, blood vessels)



•Cardiac muscle (including conduction tissue)

•Glands (eg salivary, lacrimal)

(however sensory fibers travel along with autonomic nerve fibers)

General Characteristics of ANS

Two Major Divisions: differ anatomically, in transmitters used, effect on various organs

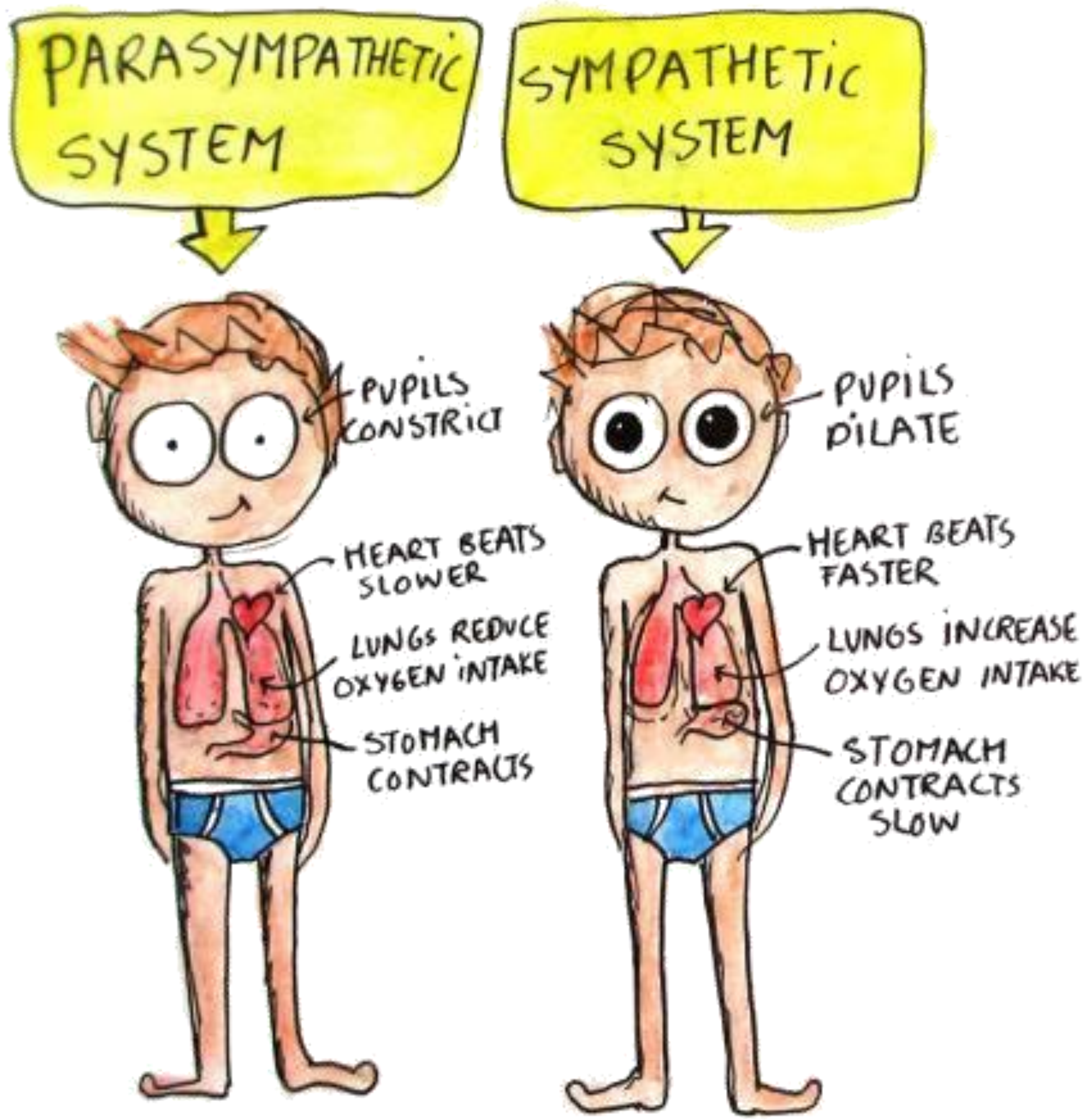
- Sympathetic (SNS): controls organs in times of stress

- Parasympathetic (PSNS): controls organs when the body is at rest

Most organs are innervated by both sympathetic and parasympathetic fibers; dual innervations results in dynamic antagonisms - they counterbalance each other's activity (In few cases, both divisions work in parallel)

- *Enteric (minor third component which is gut-related)*

Functions of the ANS



Functions of the ANS

- **Sympathetic nervous system**
  - **Fight, flight or fright mode**
  - Catabolic system
  - Increased arterial pressure
  - Increase heart rate, force of cardiac contraction and cardiac output
  - Increased blood flow away from skin to skeletal muscles - cold, sweaty skin (constrict vessels to skin and gut / dilate vessels to heart, muscles, lungs)
  - Increased ventilation – dilation on airways- deep & rapid breathing
  - Pupillary dilation (greater field of vision)
  - Sweat glands

- **Parasympathetic nervous system**
  - **Rest and digest mode**
  - Homeostatic or anabolic system
  - Decreased blood pressure
  - Decreased heart rate
  - Constrictor of bronchial tree and decreased respiratory rates
  - Stimulation of gut activities - allows for digestion and waste removal
  - Warm skin
  - Pupillary constrictor

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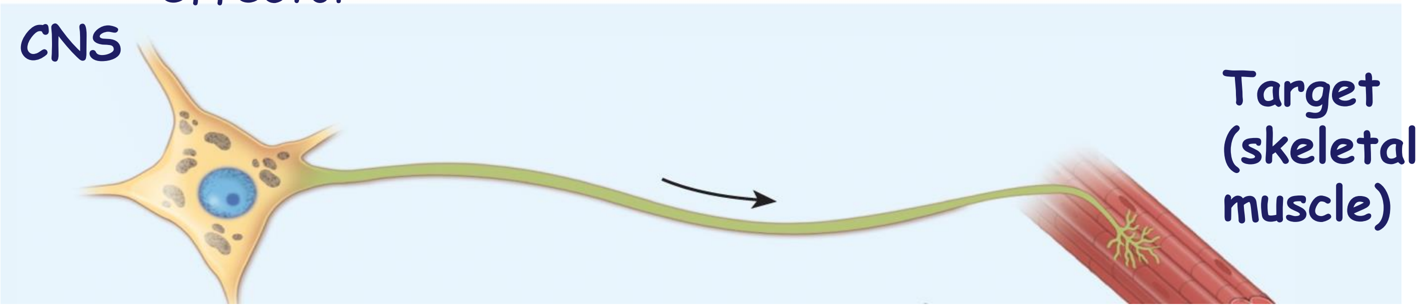
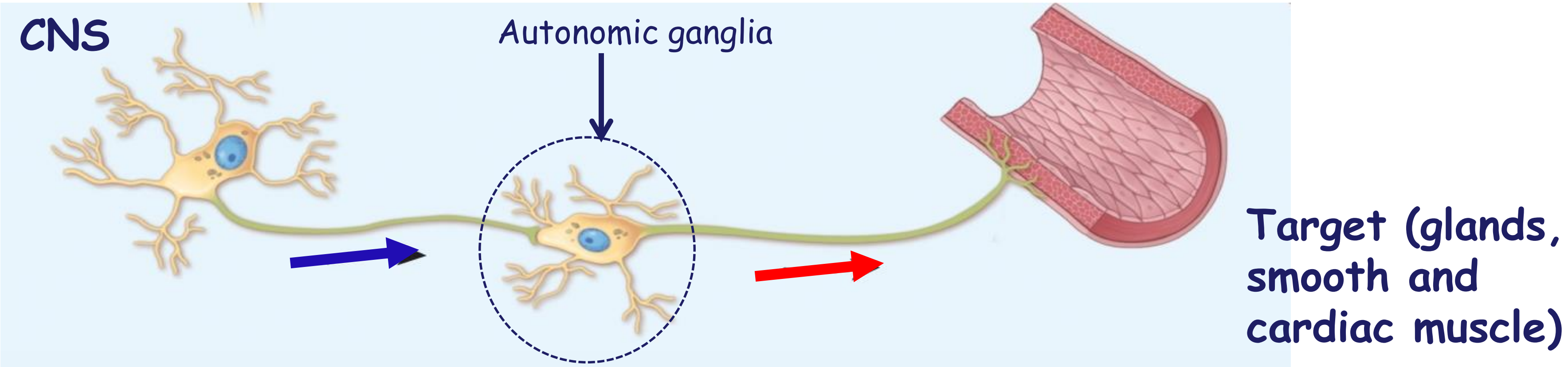
Somatic vs Autonomic Motor Pathways

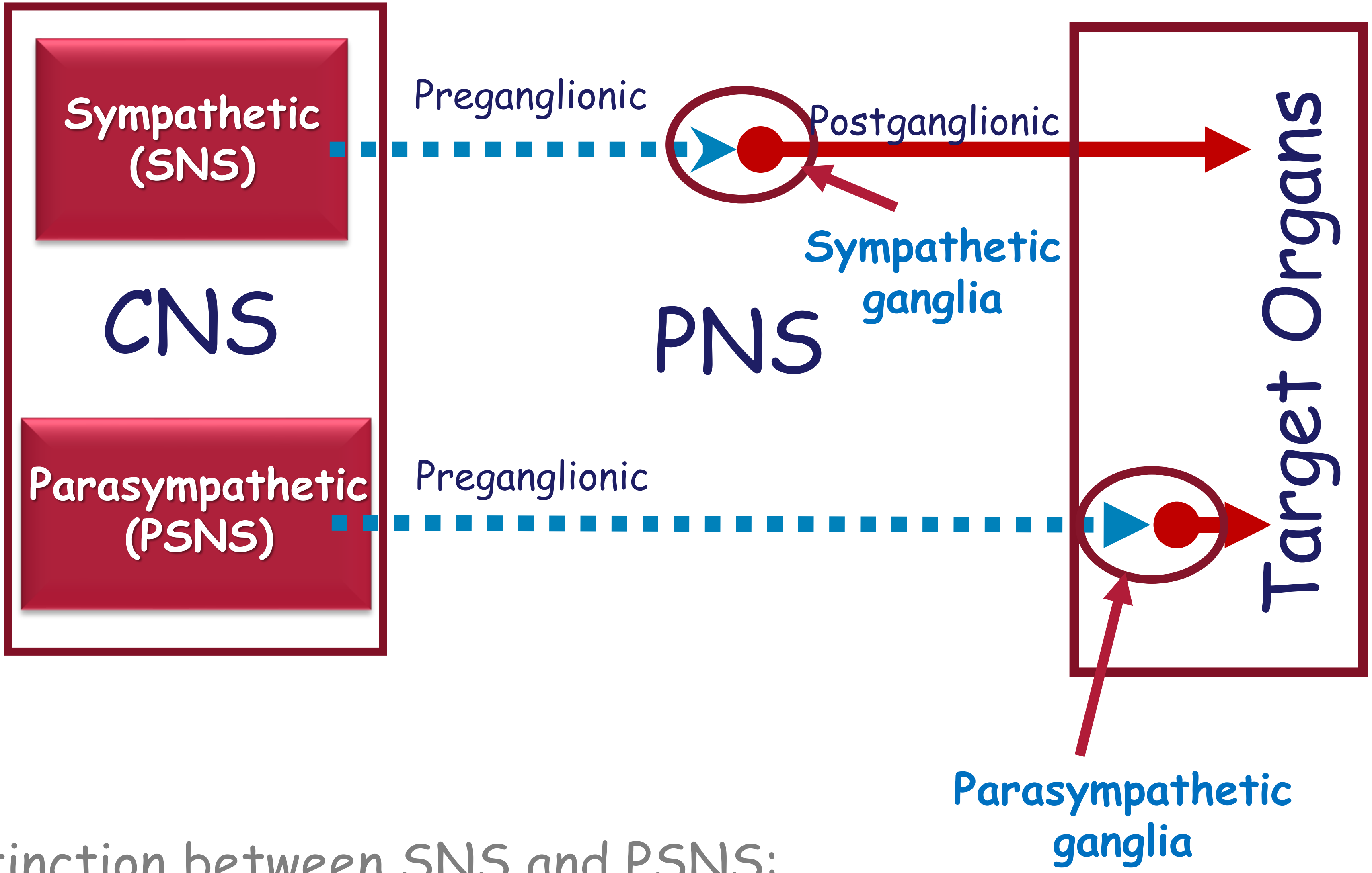
Somatic motor neurons: One long, heavily myelinated axon extending from the CNS to the effector



Autonomic motors neurons: Two Neuron System

- A lightly myelinated **preganglionic** neuron with the cell body in the CNS
- An unmyelinated **postganglionic** neuron with the cell body in autonomic ganglion in the PNS





Distinction between SNS and PSNS:

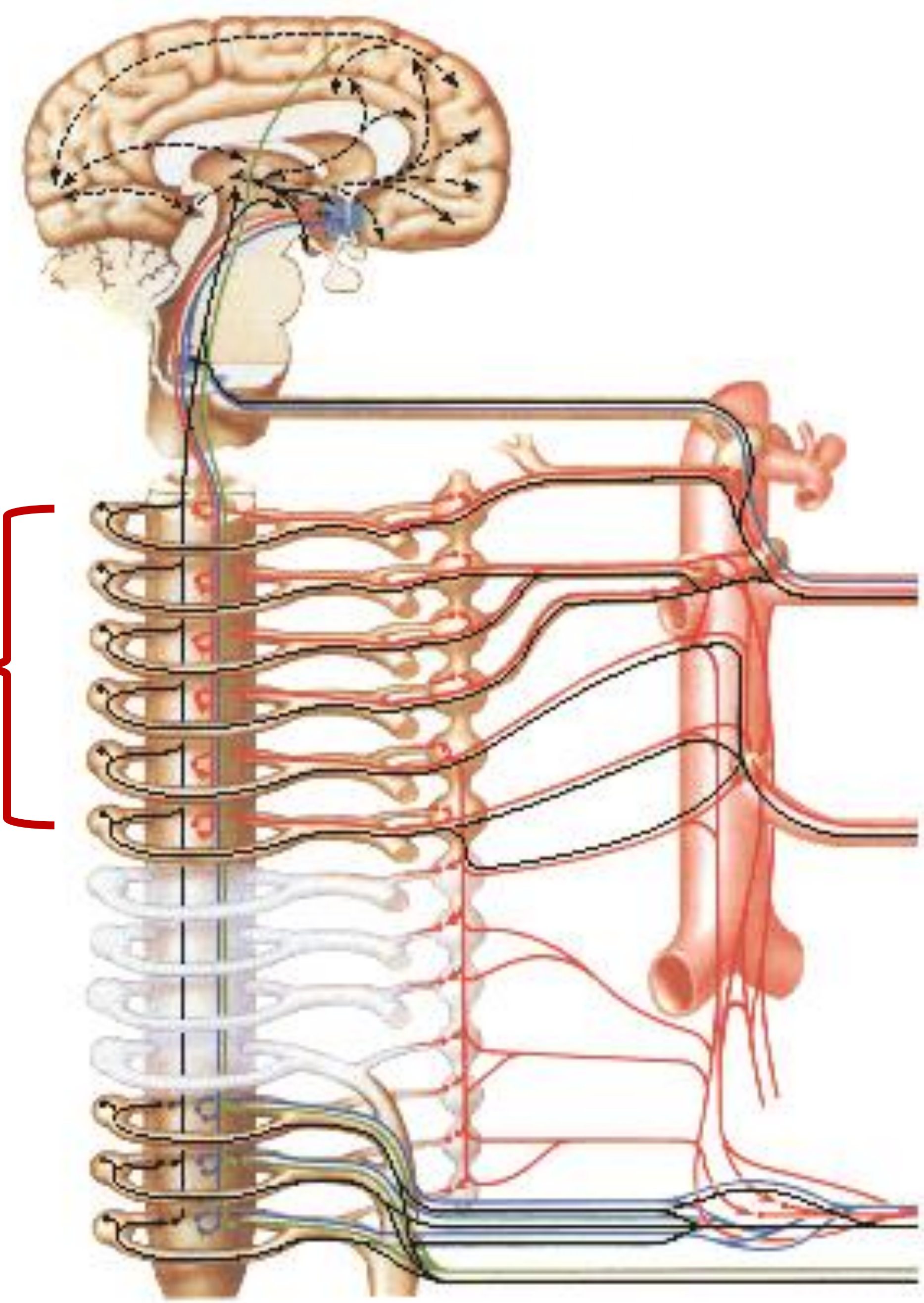
- Location of the pre-ganglionic cell bodies
- Location of the post-ganglionic cell bodies

Preganglionic Sympathetic

Where are the cell bodies located?

- Lateral horn from T1 to L2 spinal cord segments

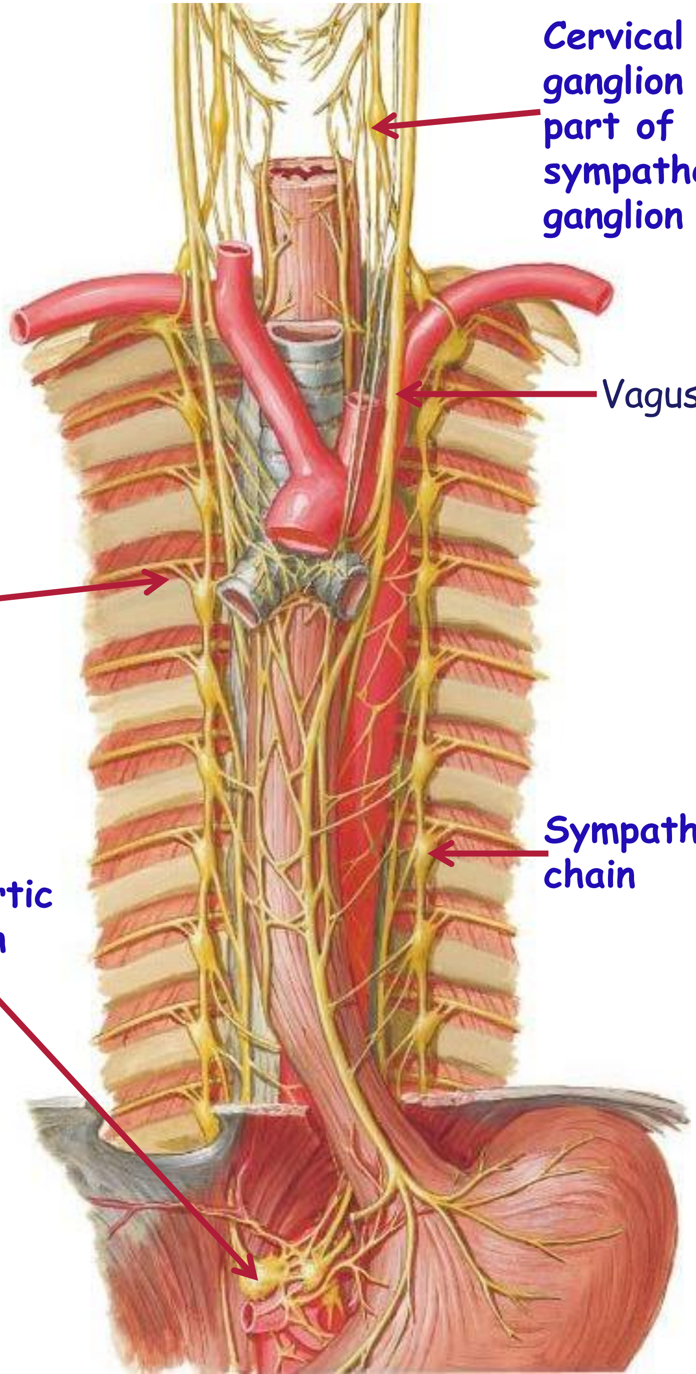
Sympathetic outflow
Thoracolumbar
T1 –L2



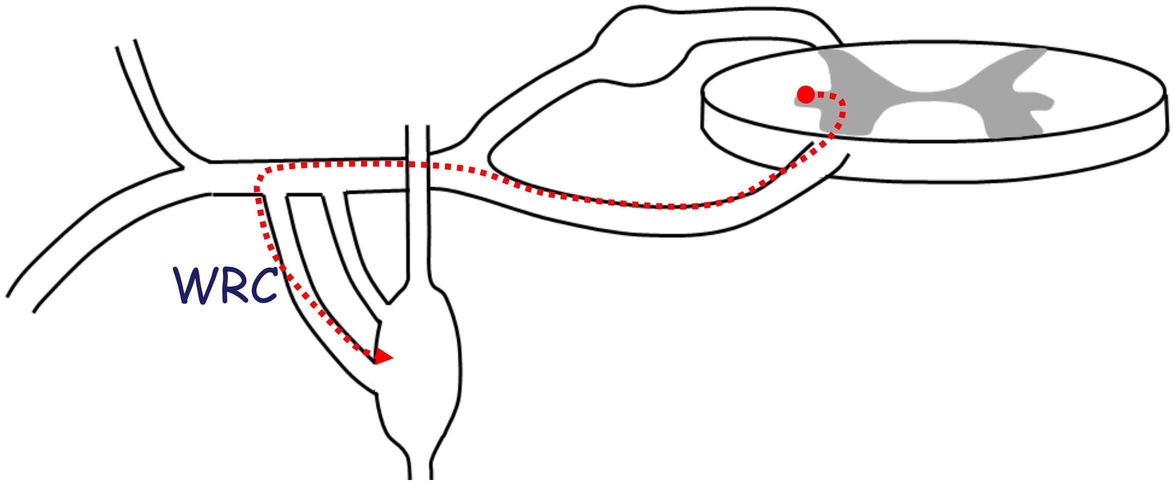
Postganglionic Sympathetic

Where are the cell bodies located?

- *Sympathetic Ganglia*
  - –SYMPATHETIC CHAIN (TRUNK) GANGLIA (paravertebral ganglia)
    - •Lateral aspect of vertebrae
    - •From superior cervical ganglion at the base of skull
    - •To the ganglion impar at the level of the coccyx
    - •**Associated with each spinal nerve: 1 pair per spinal nerve**
  - – PREAORTIC GANGLIA (prevertebral ganglia)
    - •Located in the abdomen, associated with major branches of the abdominal aorta



Sympathetics Routes



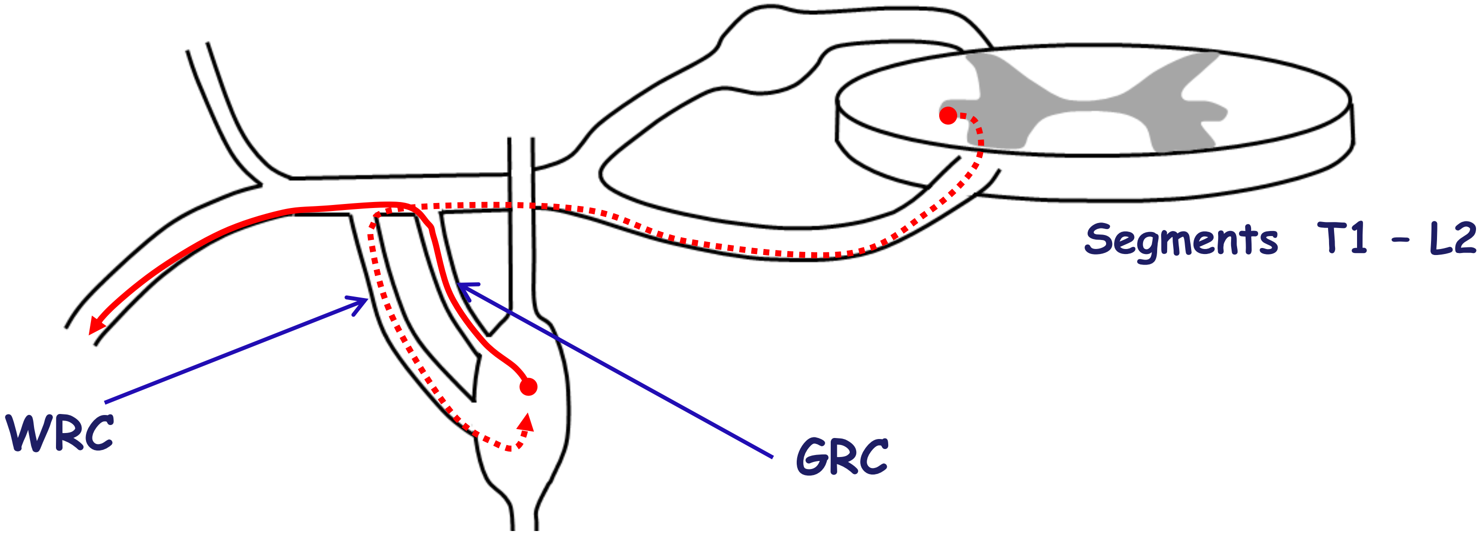
Lateral Horn T1-L2 → Ventral root → Spinal nerve/Ventral rami → WRC → Sympathetic chain

Preganglionic sympathetic nerves enter the sympathetic chain via WRCs: therefore, WRCs are only found in association with spinal nerves T1 –L2 / chain ganglia T1–L2.

From the chain, there are several routes:

1) To the body wall and limbs
2) To viscera (thoracic/abdominal/pelvic/perineum organs)
3) To organs/structures in the head

Sympathetics to Body Wall and Limbs



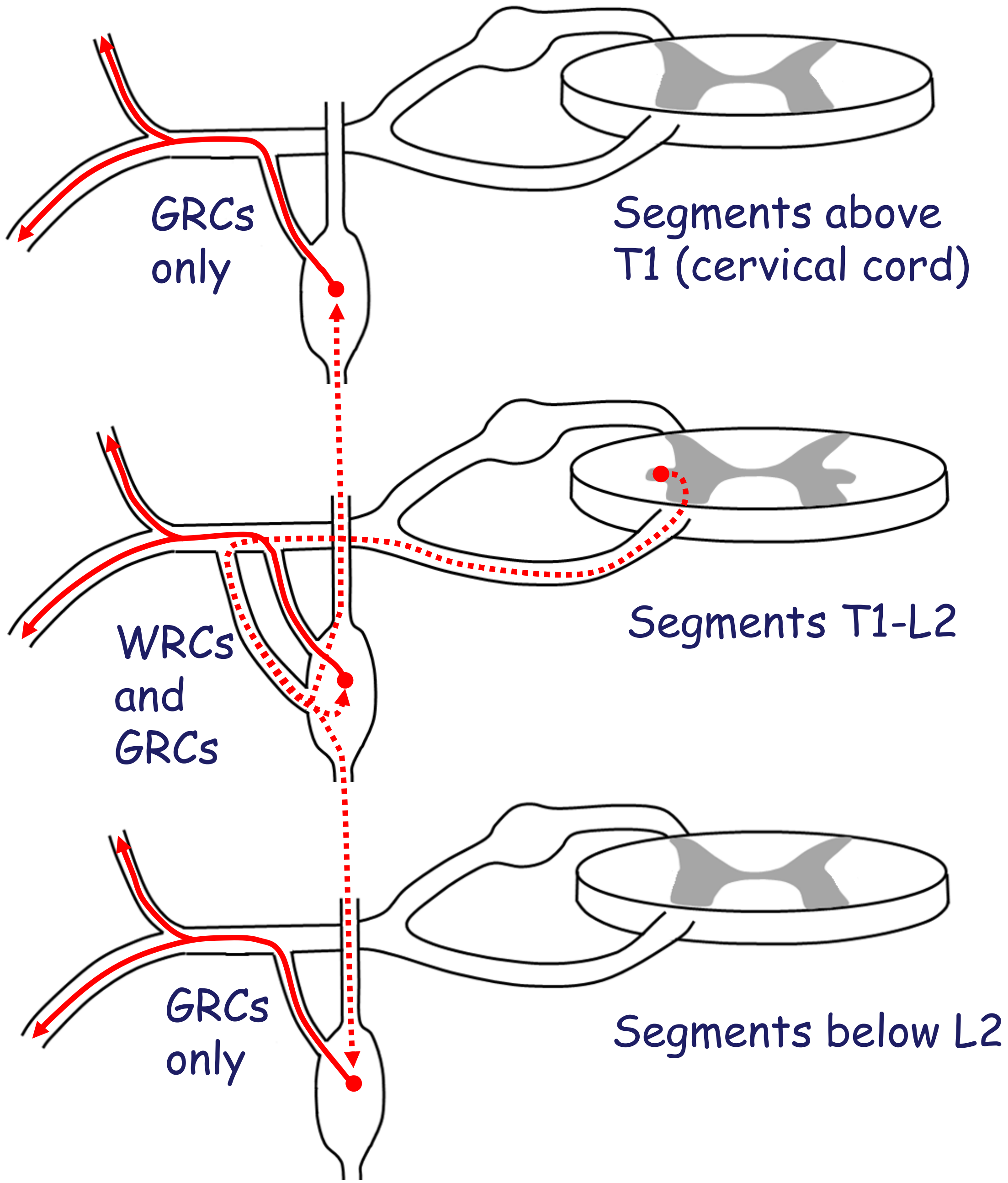
•GRCs are used by postganglionic sympathetics which are destined for the body wall and limbs

•GRCs ONLY lead to spinal nerves (the nerves of the body wall and limbs).

•GRCs are therefore found at all spinal levels.

• Above T1 and below L2 there are only GRCs.

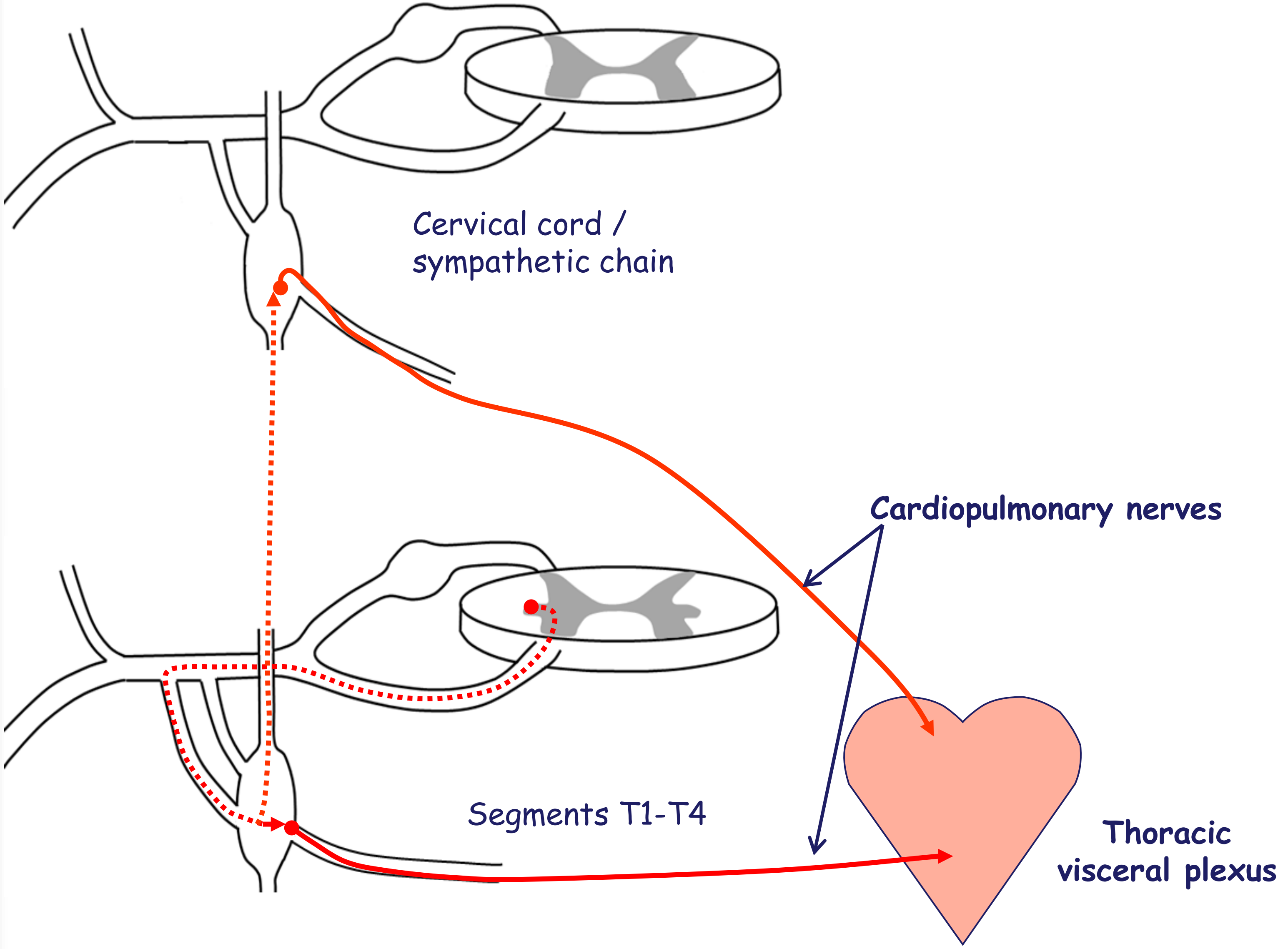
Sympathetics to Body Wall and Limbs



There are only 3 types of sympathetic targets in the body wall and limbs:

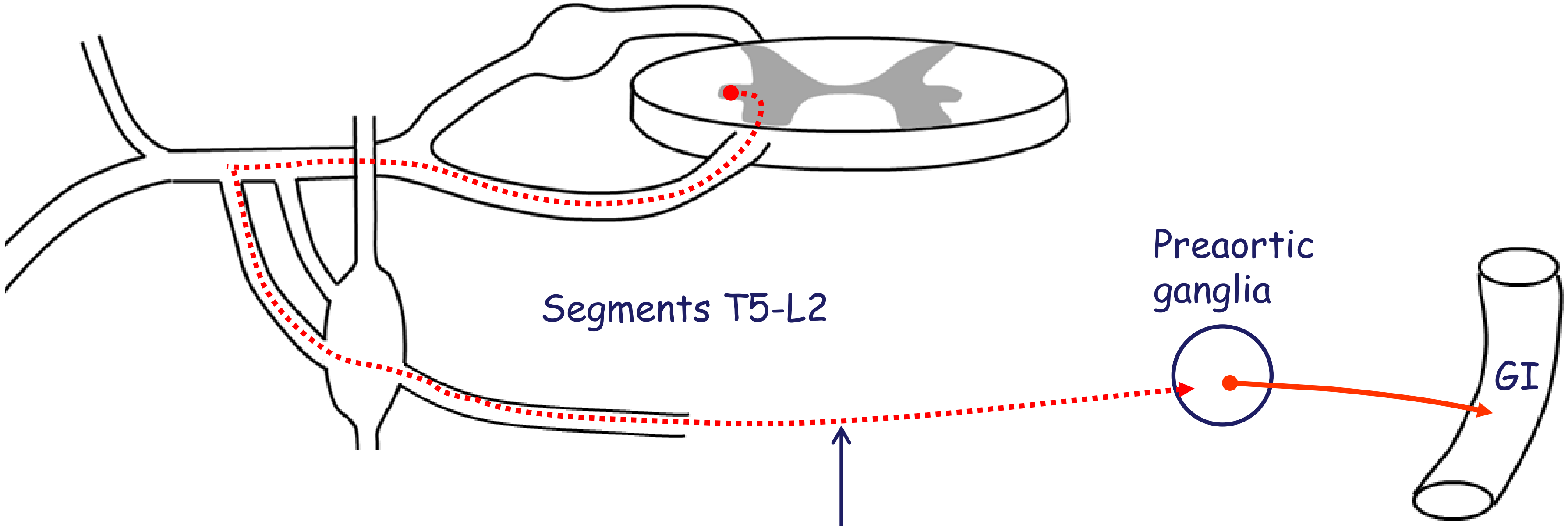
1) Sweat glands
2) Vascular smooth muscle (vessels in the skin, vessels in the connective tissues and vessels in skeletal muscles)
3) Arrector pili muscles

Sympathetics to Thoracic Organs



- Preganglionic sympathetics with cell bodies in lateral horn from T1-T4 enter chain via WRCs, *synapse in chain ganglia at the same level.*
- Postganglionic axons exit the chain medially and travel to the cardiopulmonary plexus.
- Other preganglionic sympathetics with cell bodies in lateral horn from T1-T4 enter chain via WRCs, *ascend in the chain and synapse in cervical chain ganglia.*
- Postganglionic axons exit the chain medially and travel to the cardio-pulmonary plexus.

Sympathetics to Abdominal and Pelvic Organs

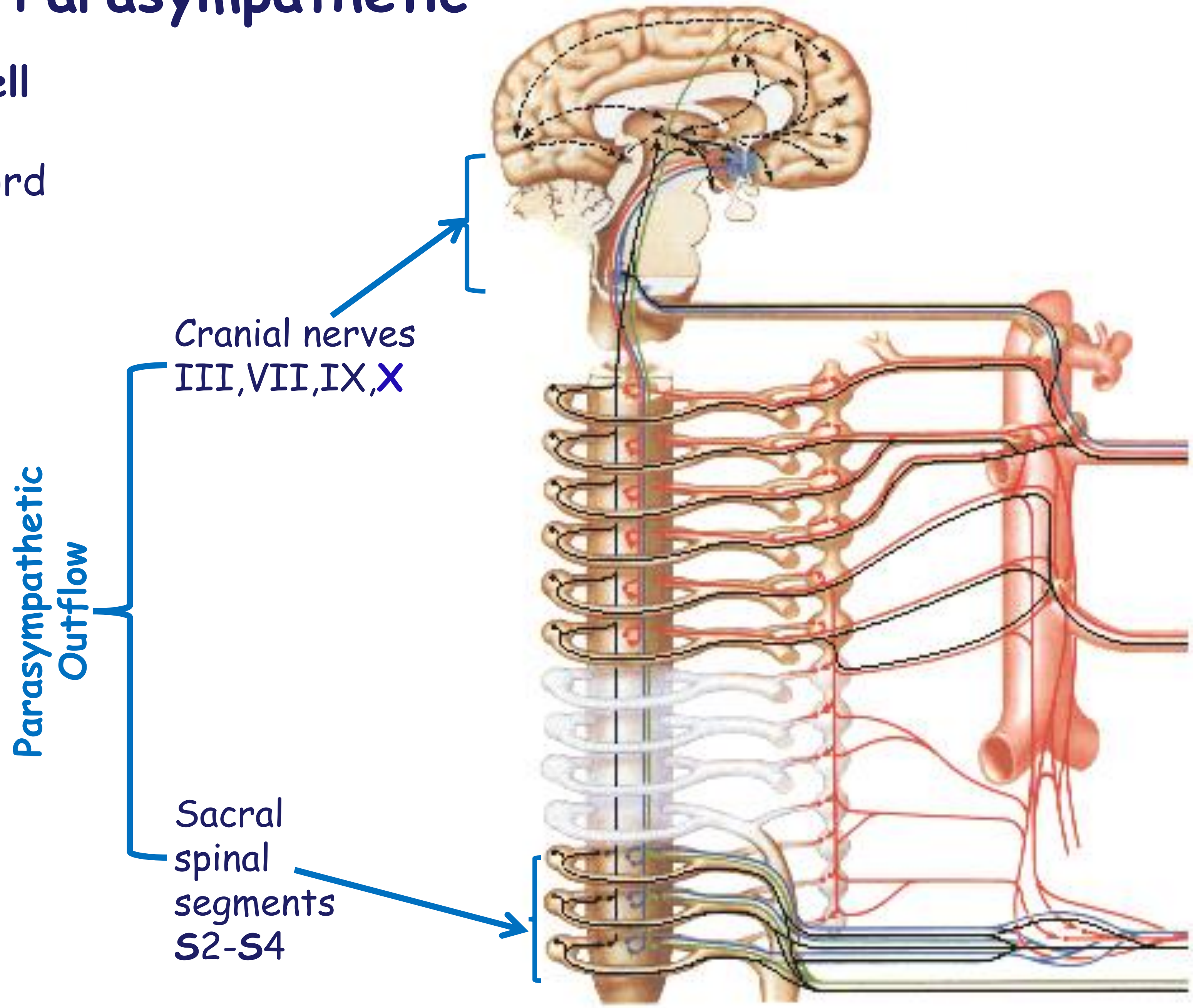


- **Preganglionic axons** with cell bodies in the lateral horn from **T5–L2** enter the chain, *pass on through ganglia **WITHOUT** synapsing.*
- The preganglionic fibers (called splanchnic nerves) then leave the medial aspect of the chain ganglia and travel to a sympathetic ganglion along the abdominal aorta (preaortic = prevertebral ganglia) where they synapse
- Postganglionic fibers then travel along arteries to the GI tract and abdominal organs
- Postganglionic fibers form hypogastric plexus to the pelvic organs

Preganglionic Parasympathetic

Where are the cell bodies located?

- S2-S4 spinal cord segments
- Cranial nerves III,VII,IX,X



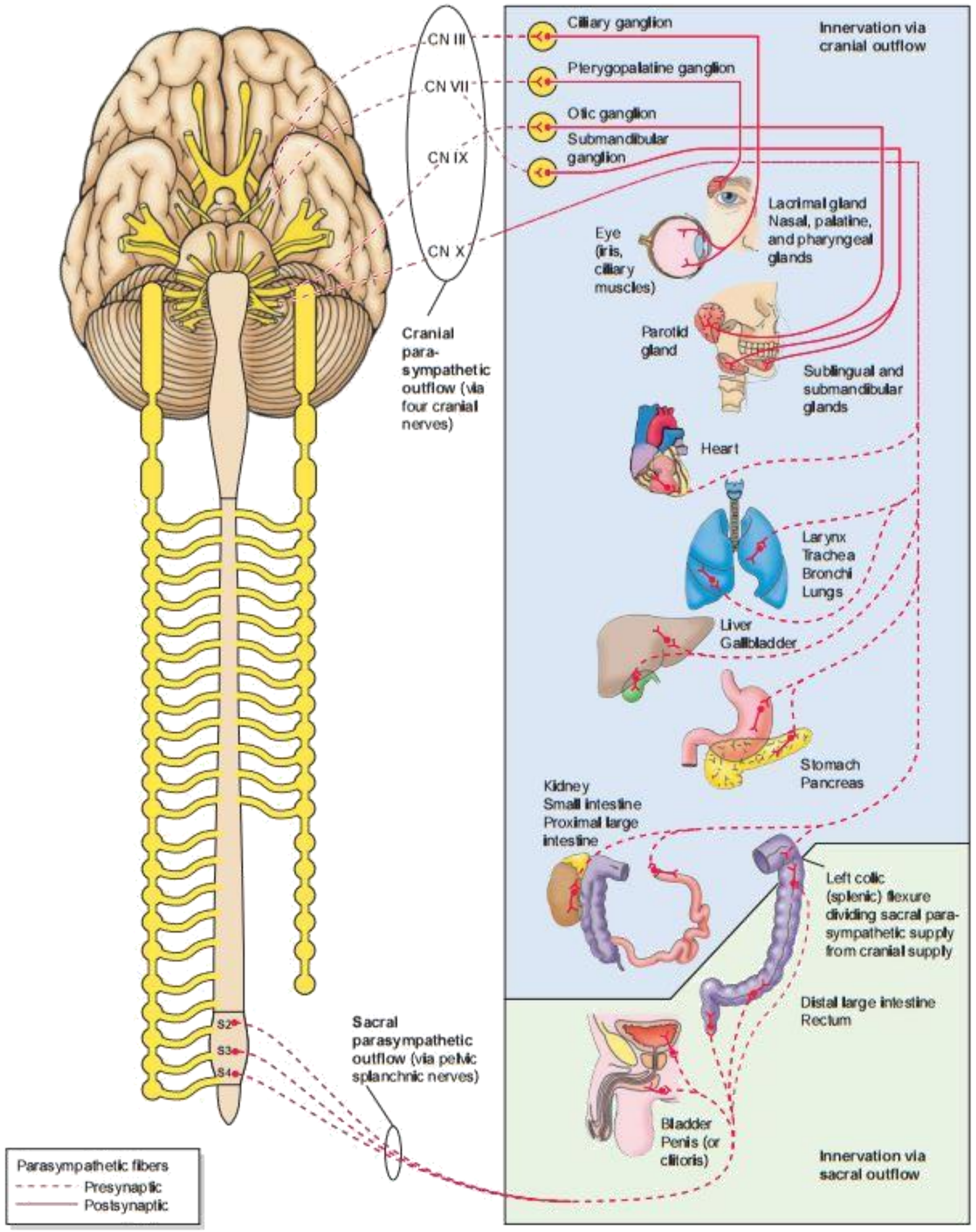
Postganglionic Parasympathetic

Where are the cell bodies located?

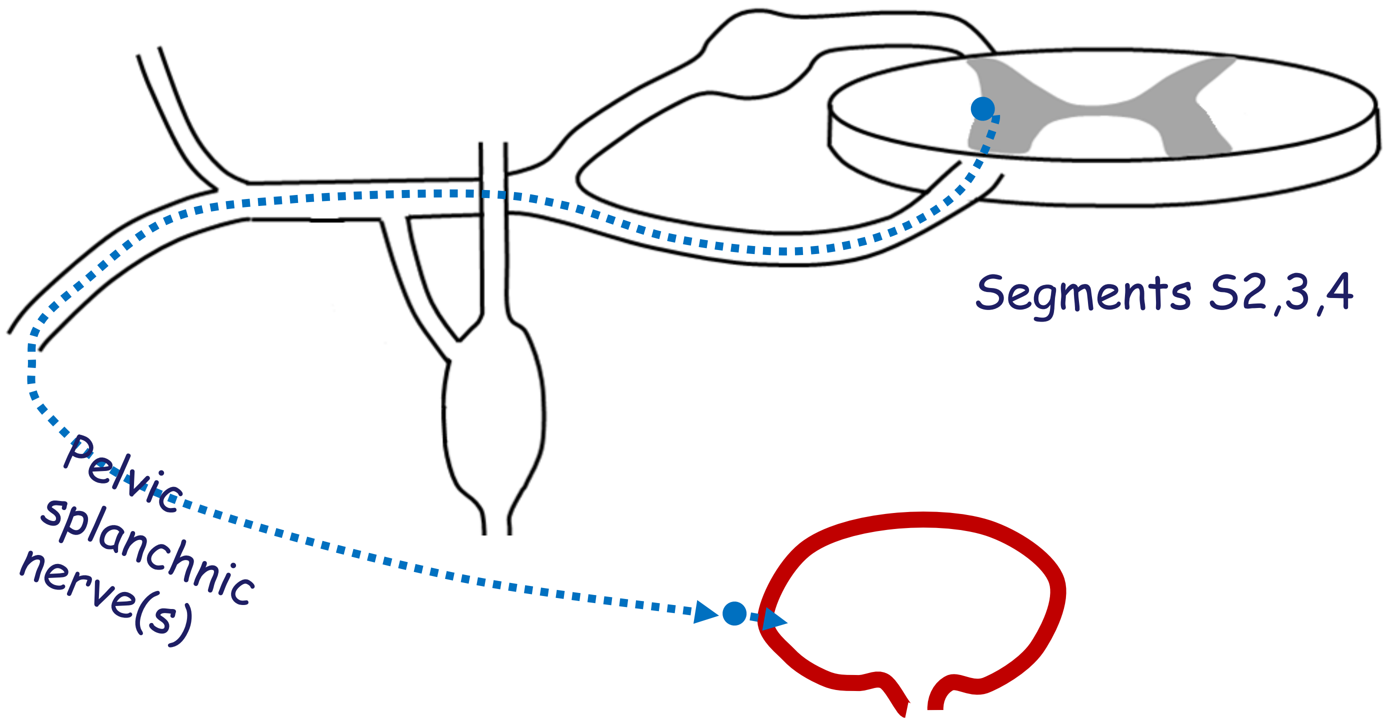
❖*Parasympathetic Ganglia*

•postganglionic parasympathetic cell bodies are found scattered near or in the walls of target organs

•discrete parasympathetic ganglia are <u>only</u> found in the head associated with the parasympathetic cranial nerves.



Parasympathetics to Body



- **Preganglionic parasympathetics:**
  - cell bodies in a small “**lateral-horn-like area**” of sacral spinal cord(S2-4)
  - axons travel on the ventral roots to spinal nerves and then ventral rami of S2-4
  - leave the ventral rami as **pelvic splanchnic nerves** to enter the **pelvic plexus**
- **Postganglionic parasympathetic** nerves are located at or in the walls of target organs in the pelvis (such as the urinary bladder)

Questions email:

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